

Liquidity Misallocation in an Over-The-Counter Market*

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Abstract

To understand the illiquidity of the over-the-counter market when dealers and traders are in long-term relationships, I develop a framework to study the endogenous liquidity distortions resulting from the profit-maximizing, screening behavior of dealers. The dealer offers the trading mechanism contingent on the aggregate history of his customers summarized by the asset allocation. The equilibrium distortion is type dependent: trade with small surplus breaks down; trade with intermediate surplus is delayed; trade with large surplus is carried out with a large bid/ask spread but without delay. The distortions become more severe when the valuation shock is frequent, the valuation dispersion is large or the matching friction to form new relationships is large. Implications on the dealer's quoting strategy and asset pricing are also discussed.

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1 Introduction

Many financial over-the-counter (OTC) markets are illiquid, with large bid-ask spreads and long delay in trade, which distorts asset prices and allocation efficiency. Standard theories of the OTC market such as Duffie, Garleanu and Pedersen (2005)[7] and Lagos and Rocheteau (2010)[21] show that liquidity distortions can arise from the search friction to locate trading counterparties. However, financial institutions typically maintain long-term relationships with each other.¹ The friction to locate trading counterparties may not be so large as to explain all the distortions. In this paper, I show that even when broker-dealers maintain long-term relationships with traders, additional liquidity distortions may arise if they do not observe traders' private valuations. In this environment, dealers screen traders by controlling the speed of trade and the transaction price and keep track of the endogenous asset allocation across his customers. Although dealers provide immediacy to traders, the monopsony power of the dealer in the OTC market leads to imperfect allocation of liquidity.

In the model, traders search and match with dealers. Similar to equilibrium search models of the labor market[25], the match is long-term but subject to breakup risks and forming new matches takes time. Each dealer is matched with a continuum of traders. By posting menus of contracts specifying trading probabilities and transaction prices for indivisible assets, dealers screen traders with heterogeneous gain from trade. In equilibrium, traders with large gain from trade value most immediacy. So, they are willing to pay a high premium relative to the market price to trade faster. Traders with intermediate gain from trade sacrifice trading speed for a lower spread. To induce traders with large gain to accept the high premium, dealers strategically exclude traders with small gain by charging a fixed spread on top of a variable spread increasing in the trading probability. Therefore, trade breaks down for those with small gains from trade. I show that these three types of distortions coexist in equilibrium under general conditions.

A theoretical contribution of the paper is to formalize and solve the dynamic programming problem of the dealer, in which the asset allocation to customers is a high-dimensional state

¹See, for example, Li and Schürhoff (2014)[24].

variable and the contract menus are the control variable. The asset allocation is slow moving because of the strategic delay of the dealer and the physical limit on the trading frequency between the dealer and his customers. This induces the dynamic interaction between the trading mechanism and the asset allocation, especially for traders with intermediate gain from trade. Traders with large gains from trade are willing to pay a premium to trade immediately. Trading faster with them also means that less of those traders remain waiting. With less traders with large gain remaining, dealers have an incentive to trade faster with traders with intermediate gain. This dynamic effect of the past trading activity on the current trading activity is related to the Coase conjecture, that a dealer competes with his future self when trading with investors and this drives his monopsony rent to zero. In this model, the asset allocation as a summary statistics of the aggregate history of customers matched with the dealer is also affected by other events in the past, such as relationship breakup, arrival of new traders and preference shocks to traders. So, the Coase conjecture does not hold exactly. For traders with intermediate gain from trade, trade does not break down but is delayed. For trade with small gain, it still breaks down as in a static screening problem.

The model links the opacity of the OTC market to market liquidity, transaction costs and asset prices. Opacity leads to heterogeneous valuation over an asset. I show that more dispersed valuations lead to more delay in trade and larger spread for all traders.

The information friction generates an endogenous transaction cost, which predicts a negative relationship between the transaction cost and the expected trading volume. This finding contrasts our results from Lagos and Rocheteau (2009)[21], where the relationship is positive, and is consistent with empirical findings from the OTC markets of corporate bonds (Edwards, Harris and Piwowar (2007)[9]).² Li and Schuerhoff (2013)[24] shows that in the municipal bonds market, investors also face a trade-off between execution speed and cost.

²A caveat is that in the empirical work, immediacy is measured by the actual trading volume. In my model, it can only be measured by the expected trading volume. So, the mapping between my model and the data is based on the assumption that the expected trading volume in the data is positively correlated with the actual trading volume.

Apart from the inventory concern mentioned in their paper, the mechanism offered in my paper could also contribute to this trade-off.

The information rent enjoyed by traders also distorts the asset price. I show that there exists an information premium in the asset price. The expected information rent for sellers increases the premium, while the expected information rent for buyers decreases the premium.

The endogenous liquidity distortion also depends on the stability of trader-dealer relationships and the matching friction. In benchmark models of the OTC market such as Duffie, Garleanu and Pedersen (2005)[7] and Lagos and Rocheteau (2009)[21], liquidity distortion arises because of search frictions to locate trading counterparties. I show that when it is hard to form new matches, dealers are also more likely to delay trade. Therefore, the strategic delay of the dealer amplifies the matching friction. On the other hand, the stability of the trader-dealer relationship, measured by the breakup rate of the relationship, may or may not induce more delay by the dealer, depending on the matching friction.

I also show that the optimal trading mechanism can be implemented by lotteries over transaction prices. So, the assumption that the dealer offers the contract menus is not restrictive, even when the dealer can only use outright sales contract. My theory gives an alternative explanation for the price dispersion in opaque markets described in Green (2007)[14].

Literature Review Since assets are durable and dealers have limited commitment to the trading mechanism they post. So, this paper is closely related to the literature on the Coase conjecture (see, for example, Gul, Sonnenschein and Wilson (1986)[17]). Recent developments in this literature include Fuchs and Skrzypacz (2010)[11] and Garrett (2013)[12]. My model allows changing valuations and arrival of new buyers over time, as well as the frictions in forming and maintaining long-term relationships, none of which are trivial additions to the baseline setup. The literature studies the optimal trading mechanism contingent on the individual history. By summarizing the aggregate history of the dealer's customers, which

includes trading activities and other events in the past, by the asset allocation, the trading mechanism in the paper becomes more tractable and captures the effect of not only the trading delay but also other frictions on the liquidity distortion in the market.

My paper is also related to the literature on strategic delay induced by asymmetric information. Guerrieri, Shimer and Wright (2010)[16] and Chang (2011)[4] combine directed search friction and asymmetric information. In these papers, delay in trade serves as a signaling device. One common issue with the signaling equilibrium is that it is sensitive to perturbation. For an environment with nearly complete information, the equilibrium allocation is the same as an environment with severe adverse selection as long as the support of the distribution remains the same. In this model, delay in trade serves as a screening device. Dealers take into account the effect of the type distribution on their profit when they optimize the menu of contracts they offer. A perturbation to the distribution will not affect qualitatively the equilibrium allocation. In this sense, the equilibrium allocation is more robust in this model. Chiu and Koeppl (2011)[6], Camargo and Lester (2011)[3], Guerrieri and Shimer (2011)[15] and Chang (2011)[4] study the effect of lemons problem on liquidity. These papers take the sales contract as given, while here I allow contracts consisting of two ingredients, the trading probability and the transaction price for sales.

Another related literature is about asymmetric information in the decentralized dynamic market. Lauermann and Wolinsky (2011)[22], Wolinsky (1990)[26] and Blouin and Serrano (2001)[1] study information aggregation in the lemon market and its social welfare as the friction vanishes. Golosov, Lorenzoni, and Tsyvinski (2009)[13] and Camargo and Lester (2011)[3] are concerned with the trading dynamics. Horner and Vieille (2009)[19] studies the interaction of strategies of sequentially arriving short-lived buyers in a dynamic lemon market. Inderst (2005)[20] studies limiting property of a matching market with adverse selection. Hendel and Lizzeri (1999)[18] studies the durable goods market with lemons. These papers study the implications of asymmetric information about the quality of the goods or assets being traded, while this paper studies the implications of asymmetric information about private valuations. Most of these papers focus on the limiting results as frictions

are asymptotically zero. My focus is to characterize the equilibrium with frictions to study the resulting liquidity misallocation. In Duffie, Malamud and Manso (2014)[8], the authors characterize analytically the dynamics of information aggregation in segmented markets. I do not allow information propagation in the model. So valuations across traders do not interact. While my focus is on the liquidity distortion due to the interaction between asymmetric information and other frictions and shocks in the economy, it would be interesting to further study the interaction between these two channels: information aggregation and the liquidity distortion that arises from the dealer's screening strategies.

The rest of the paper is organized as follows. In Section 2, I lay out the model. In Section 3, I characterize the equilibrium. In Section 4 and 5, I focus on distinct implications of the model on market liquidity, the quoting strategy and the asset price.

2 Model

Environment, Endowment and Preferences The economy is set in continuous time and lasts forever. There is a continuum of long-lived traders and dealers. The measures of both groups of agents are normalized to one. Traders are endowed with one type of asset and deep pockets of numeraire goods. An asset bears a unit flow of dividend goods. The total asset supply is $A \in (0, 1)$. The asset is homogeneous in quality. I assume that traders can hold either zero or one unit of the asset.³

Traders' valuation over the dividend good depends on their preference type x , which is her private information.⁴ A trader with preference type x enjoys flow utility x from holding one unit of the asset. Over time, a trader may experience a preference shock, which arrives with Poisson rate δ . After the shock, she draws randomly a new preference type from a continuous distribution $G(\cdot)$ on an interval $[x_L, x_H] \subseteq \mathbb{R}_+$. Both the arrival time of a preference shock

³This assumption makes the distribution of asset holdings more tractable.

⁴The asset holding can also be thought of as unobservable. Since holding more than one unit of asset or selling more than the trader's holding does not generate any profit, it is better off for traders to reveal truthfully their asset holding.

and the realization of the shock are idiosyncratic. Dealers do not value dividend goods. All agents are risk neutral and derive one util from consuming a numeraire good. Traders and dealers share a common discount rate, $r \in \mathbb{R}_+$.

The Structure of the OTC Market Agents trade assets for numeraire goods through an over-the-counter market. The OTC market has two tiers as in Lagos and Rocheteau (2009)[21]. Only dealers can trade in the first tier of the market, which we call the inter-dealer market. The inter-dealer market is competitive, where dealers buy and sell any amount of assets at the market price without delay. Denote the market clearing price at t to be P_t . The second tier is a bilateral market between dealers and traders. Dealers and traders match randomly. A trader can be matched with only one dealer. But the dealer is not constrained in his capacity to host customers. An unmatched trader meets a dealer with Poisson rate α . The relationship between a matched trader and her dealer breaks up at Poisson rate γ . Therefore, the trader-dealer relationship is long-term. This market structure can be thought as the stable core-periphery structure observed in OTC markets.⁵

Trade between a dealer and his customers is bilateral and takes place repeatedly in each dealer-customer relationship at Poisson rate q . The arrival rate represents the speed limit at which the dealer can process a trade, which includes the time to bargain and deliver the asset. Also, the dealer may be busy handling trades from his other customers, so each trader has to wait for her turn. Only when the trader finds the occasion that the dealer is not busy, he gets to negotiate the terms of the bilateral trade. This is also part of the physical delay represented by the Poisson rate.

On each trading occasion, the dealer offers two feasible menus of contracts, a menu of bid contracts for traders who want to sell and a menu of ask contracts for traders who want to buy. The trader chooses one contract from the menus.

Trading Mechanism Posted by Dealers Each contract in menu i has two components, the probability of trade, denoted by $q_{imt}/q \in [0, 1]$, and the transaction price, denoted by

⁵See, for example, Li and Schürhoff (2013)[24].

$p_{imt} \in \mathbb{R}_+$, where $i \in \{a, b\}$ indicates whether it is an asking contract or a bidding contract and $m \in \mathbb{X}$ indicates the reported preference type of the corresponding customer. A contract menu $i \in \{a, b\}$, denoted by $\mathbb{M}_{it} = \{(q_{imt}, p_{imt})\}_{m \in \mathbb{X}}$, is *feasible* if (1) given the menu, it is incentive compatible for traders to report truthfully their preference types; (2) it is individually rational for a trader of type x to accept ex ante a contract with index x ; (3) it is individually rational for a trader to trade at price p_{ixt} ex post.

The trading mechanism offered by the dealer is contingent on the *aggregate history* for the group of customers matched with him, summarized by the asset allocation to his trader customers. The summary statistics for the group of traders is analog to the posterior belief about the preference type of a trader as the summary statistics for the individual history, if the mechanism were contingent on the individual history. If the dealer trades faster with traders of a certain type before t , there will be less trading request from traders of that type at t . The summary statistics also includes other historical information, such as the arrival of new traders, the breakup of existing relationships and the arrival of preference shocks.

If the dealer has perfect information about the history of a customer, the contract menu should be further contingent on the history. In the model, I focus on the aggregate history for the following reasons. (1) While dealers can observe the history of a trading account, a customer can hide his history by opening several accounts with the same dealer at negligible cost. She can also hide the timing at which she experiences a preference shock. These hidden actions prevent the dealer from observing the history of his customers. (2) Trading mechanisms contingent on the history of each customer, which includes not only the trading history but also the history of preference changes, the date of starting the long-term relationship and the prior of her preference type, may require too much sophistication, especially because a dealer hosts a continuum of customers.

Although the ex post trading volume is always one or zero, the model has a notion of heterogeneity in trading volume. The trading probability, q_{imt}/q , is proportional to the expected trading volume for the trader⁶, while avoiding tracking the distribution of continuous

⁶The expected trading volume before breakup or a preference shock is $\frac{q_{imt}}{q_{imt} + \gamma + \delta}$.

asset holding.

The contract menus can be viewed as a direct mechanism, which can have alternative implementations where traders do not have to report explicitly their types. One implementation is to use price lotteries on each trading occasion, which fits well the institutional detail of brokers or market makers intermediating trade in the OTC market. I will show in Section 5.1 that by posting quotes randomly according to optimally chosen distributions, dealers can still the direct trading mechanism.

3 Equilibrium Definition and Existence

3.1 Equilibrium Definition

In this section, I will first present the problems of the trader and the dealer, followed by other equilibrium conditions. The equilibrium definition follows in the end. I will focus on the symmetric equilibrium.

Traders' Problem Denote the expected payoff of a trader of preference type x with a units of asset and matched with a dealer at t to be M_{axt} , where $a \in \{0, 1\}$. Denote the expected payoff of a trader with preference type x with a units of asset and searching for a dealer at t to be U_{axt} . M_{1xt} is characterized by the following Hamilton-Jacobi-Bellman (HJB) equation,

$$\begin{aligned} & \left(r + \int_{x_L}^{x_H} q_{bmt} ds_{1xt}(m) + \gamma + \delta \right) M_{1xt} \\ &= x + \max_{s_{1xt}} \int_{x_L}^{x_H} q_{bmt} (M_{0xt} + p_{bmt}) ds_{1xt}(m) + \gamma U_{1xt} + \delta E M_{1t} + \dot{M}_{1xt}. \end{aligned} \quad (1)$$

The payoff depends on the contract choice of the trader and her continuation value after the customer-dealer relationship breaks up, preference shocks and transactions. s_{1xt} denotes the contract choice over the menu of bid contracts, which is a probability distribution over the contracts in menu $\mathbb{M}_{bt}$, $s_{1xt} \in \Delta([x_L, x_H])$. The contract choices are made at frequency q . And q_{bmt}/q is the probability of trade conditional on choosing the contract with index m .

The continuation value after selling the asset is $M_{0xt} + p_{bmt}$. The continuation value after a breakup shock, which arrives at rate γ , is U_{1xt} . The continuation value of a preference shock, which arrives at rate δ , is $EM_{1t} = \int M_{1\tilde{x}t} dG(\tilde{x})$.

If the contract menu, $\mathbb{M}_{bt}$, is feasible, the trader must have the incentive to choose the contract with index x , which requires

$$x \in \arg \max_{m \in [x_L, x_H]} \frac{q_{bmt}}{q} (M_{0xt} + p_{bmt} - M_{1xt}), \quad (2)$$

$$0 \leq \frac{q_{bxt}}{q} (M_{0xt} + p_{bxt} - M_{1xt}), \quad (3)$$

$$0 \leq M_{0xt} + p_{bxt} - M_{1xt} \quad (4)$$

where equation (2) is the incentive compatibility (IC) constraint for the trader, equation (3) is the ex ante individual rationality (IR) constraint and equation (4) is the ex post IR constraint. With the feasible contract menu, the HJB equation for the payoff M_{1xt} , equation (1), can be simplified as

$$(r + q_{bxt} + \gamma + \delta) M_{1xt} = x + q_{bxt} (M_{0xt} + p_{bxt}) + \gamma U_{1xt} + \delta EM_{1t} + \dot{M}_{1xt}. \quad (5)$$

Likewise, the payoff of a trader of preference type x without asset and matched with a dealer, M_{0xt} , solves the following HJB equation,

$$(r + q_{axt} + \gamma + \delta) M_{0xt} = q_{axt} (M_{1xt} - p_{axt}) + \gamma U_{0xt} + \delta EM_{0t} + \dot{M}_{0xt}, \quad (6)$$

where $EM_{0t} = \int M_{0\tilde{x}t} dG(\tilde{x})$ denotes the continuation value of the trader following a preference shock.

The continuation values of traders not matched with a dealer, U_{1xt} and U_{0xt} , solve the following equations,

$$(r + \alpha + \delta) U_{1xt} = x + \alpha M_{1xt} + \delta EU_{1t} + \dot{U}_{1xt}, \quad (7)$$

$$(r + \alpha + \delta) U_{0xt} = \alpha M_{0xt} + \delta EU_{0t} + \dot{U}_{0xt}. \quad (8)$$

The right hand side of the HJB equation for value U_{1xt} , equation (7), depends on two events. With rate α , the trader meets a market maker, his continuation value in this case is M_{1xt} .

With rate δ , the trader receives a preference shock, the continuation value in this case is $EU_{1t} = \int U_{1xt}dG(x)$. The HJB equation for value U_{0xt} , equation (7), depends on similar contingencies.

Two more value functions useful for the presentation are the valuation over an asset for traders matched with a dealer, $d_{xt} = M_{1xt} - M_{0xt}$ and for traders not matched with a dealer, $e_{xt} = U_{1xt} - U_{0xt}$. The corresponding HJB equations are displayed in the beginning of the Appendix.

Laws of Motion Denote the population density function of traders matched with a dealer to be n_{axt}^m and the density function of those not matched to be n_{axt}^u , where $a \in \{0, 1\}$. First, the dealer finding rate, α , and the relationship breakup rate, γ , determine the measure of traders matched or not matched with dealers,⁷

$$n_{0xt}^m + n_{1xt}^m = \frac{\alpha}{\alpha + \gamma}g(x), \quad (9)$$

$$n_{0xt}^u + n_{1xt}^u = \frac{\gamma}{\alpha + \gamma}g(x). \quad (10)$$

The asset allocation for traders matched with a dealer and traders not matched depends on the equilibrium contract menus offered by dealers. Denote the population density of type x traders assigned an asset to be n_{1xt}^m for those matched with a dealer and n_{1xt}^u for those not matched. The laws of motion for these two population density functions are

$$\dot{n}_{1xt}^m = \alpha n_{1xt}^u - \delta n_{1xt}^m + \delta g(x)N_1^m - \gamma n_{1xt}^m - q_{bxt}n_{1xt}^m + q_{axt}n_{0xt}^m, \quad (11)$$

$$\dot{n}_{1xt}^u = -\alpha n_{1xt}^u - \delta n_{1xt}^u + \delta g(x)N_1^u + \gamma n_{1xt}^m, \quad (12)$$

where $N_a^k \equiv \int n_{axt}^k dx$ denotes the total measure of traders with a unit of asset and matching status $k \in \{m, u\}$. According to equations (9) and (10), $N_1^m = \frac{\alpha}{\alpha + \gamma}A$ and $N_1^u = \frac{\gamma}{\alpha + \gamma}A$. Apart from the inflows and outflows induced by exogenous events such as matching or break up with a dealer, and receiving a preference shock, the trading frequencies picked by dealers, q_{bxt} and q_{axt} , affect asset allocation for traders matched with a dealer. The delay in trade

⁷This is because $\frac{d}{dt}(n_{0xt}^m + n_{1xt}^m) = \alpha(n_{0xt}^u + n_{1xt}^u) - \gamma(n_{0xt}^m + n_{1xt}^m)$ and $n_{0xt}^u + n_{1xt}^u + n_{0xt}^m + n_{1xt}^m = g(x)$.

specified in dealers' contract menus also imposes an externality on the asset allocation to unmatched traders. This affects the strategic interaction across dealers because dealers take the inflow from unmatched traders as given.

Dealers' Problem A dealer's expected payoff at t , denoted by $D_t(n_1^m)$, depends on the expected profit from each contract and the distribution of contract choices by his customers. Asset allocation to his customers, represented by n_1^m , is the state variable for the dealer's value function. The state variable summarizes the history of his customers. It also includes other historical information, such as the arrival of new traders, the breakup of existing relationships and the arrival of preference shocks. It is an endogenous state variable. If the dealer sells assets to traders of type x at a higher frequency, more assets will be allocated to type x traders. If the dealer buys assets at a higher frequency, less assets will be allocated to them. The asset allocation in turn affects the equilibrium demand and supply of assets from traders and therefore affects the dealer's incentive to screen traders. More explicitly, the value function is solved by the following HJB equation,

$$rD_t(n_1^m) = \max_{\mathbb{M}_a, \mathbb{M}_b} \int_{x_L}^{x_H} [q_{ax}(p_{ax} - P_t)n_{0x}^m + q_{bx}(P_t - p_{bx})n_{1x}^m + f(x)\frac{\partial D_t(n_1^m)}{\partial n_{1x}^m}]dx + \dot{D}_t(n_1^m), \quad (13)$$

subject to, $\forall x \in [x_L, x_H]$,

$$f(x) = \alpha n_{1xt}^u - \delta n_{1x}^m + \delta g(x)N_1^m - \gamma n_{1x}^m - q_{bx}n_{1x}^m + q_{ax}n_{0x}^m, \quad (14)$$

$$n_{0x}^m = \frac{\alpha}{\alpha + \gamma}g(x) - n_{1x}^m, \quad (15)$$

$\mathbb{M}_a, \mathbb{M}_b$ are feasible.

In the objective function of the profit maximizing dealer, $f(x)\frac{\partial D_t(n_1^m)}{\partial n_{1x}^m}$ is a key term that differentiates dynamic screening from static screening. $f(x) = \dot{n}_{1x}^m$ is solved by equation (14), the law of motion of the asset allocation function, $n_{1x}^m \cdot \frac{\partial D_t(n_1^m)}{\partial n_{1x}^m}$ is the marginal value of the asset allocation to type x traders. The product of $f(x)$ and $\frac{\partial D_t(n_1^m)}{\partial n_{1x}^m}$ represents the

marginal effect of changing asset allocation to type x traders on the continuation value of the dealer. More analysis on this dynamic effect will follow in equilibrium characterization.

Market Clearing Condition The flow demand and supply of the asset in the inter-dealer market at any moment is equal to the bid and ask transaction flow respectively.

$$\int q_{axt} n_{0xt}^m dx = \int q_{bxt} n_{1xt}^m dx \quad (16)$$

Definition 1. Given the initial distribution of traders, a symmetric equilibrium is value functions, $\{M_{axt}, U_{axt}, D_t(\cdot)\}$, laws of motion for population distribution $\{n_{axt}^m, n_{axt}^u\}$ and traders' contract choices $\{s_{axt}\}$ for all $a \in \{0, 1\}, x \in \mathbb{X}, t \in \mathbb{R}_+$, contract menus, $\mathbb{M}_{at}$ and $\mathbb{M}_{bt}$ for all $t \in \mathbb{R}_+$, and market prices $\{P_t\}_{t \in \mathbb{R}_+}$ of the inter-dealer market such that for all $a \in \{0, 1\}, x \in \mathbb{X}, \tau \geq t$,

- given the value functions of unmatched traders $U_{ax\tau}$, the value functions of matched traders $M_{ax\tau}$ and their contract choices $s_{ax\tau}(m) = \mathbb{I}\{m \geq x\}$ solve traders' problem;
- given the value functions of matched traders, the value functions of unmatched traders solve equations (7) and (8);
- given $n_{ax\tau}^u$ and $U_{ax\tau}$, the contract menus $\mathbb{M}_{a\tau}, \mathbb{M}_{b\tau}$ and distribution $n_{ax\tau}^m$ solve dealers' problem at period t ;
- given the equilibrium contract menus, $\{n_{ax\tau}^m, n_{ax\tau}^u\}$ satisfy the laws of motion specified in equations (9), (10), (11) and (12);
- the inter-dealer market clears at τ , with the market clearing condition specified in equation (16).

3.2 Characterization of Stationary Equilibrium

In a stationary equilibrium, $\dot{n}_{axt}^j = 0, \forall i \in \{0, 1\}, j \in \{m, u\}, \dot{M}_{1xt} = \dot{M}_{0xt} = \dot{U}_{1xt} = \dot{U}_{0xt} = \dot{D}_t = 0, \forall x, t$ and the equilibrium contract menus are not time varying. So we omit the time index in the notation.

The key to characterizing the equilibrium is to solve dealers' dynamic screening problem. To simplify the problem, we first separate it into two sub-problems, optimizing bid contracts and optimizing ask contracts. This is possible because there exists a cutoff preference type, κ , such that: (1) for traders with preference type above κ , the gain from selling assets is negative even if the trade is carried out at the competitive inter-dealer market price without any delay; (2) for traders with preference type below κ , the gain from buying assets is negative.

Assumption 1. *The interdealer market price $P \in [d_{x_L}, d_{x_H}]$. Trading frequency functions, $q_a : [x_L, x_H] \rightarrow [0, q]$ and $q_b : [x_L, x_H] \rightarrow [0, q]$, specified in equilibrium contract menus are continuous almost everywhere.*

Assumption 1 is a common assumption in the lemmas for Theorem 1. It specifies some equilibrium conditions that can be verified in the proof for the theorem.

Lemma 1. *Under Assumption 1, d_x is continuous, differentiable almost everywhere and strictly increasing in x . And there exists a unique cutoff κ such that only types $x \geq \kappa$ are willing to buy and types $x \leq \kappa$ are willing to sell. So $q_{bx} = 0$, for all $x \geq \kappa$ and $q_{ax} = 0$, for all $x \leq \kappa$. At $x = \kappa$, d_κ is equal to the asset price P .*

The proof of the lemma is in Section A.1 of the Appendix. According to the lemma, dealers can solve for the bid contract menu and the ask contract menu in two separate problems, taking the cutoff preference type, κ , as given.

The next step is to simplify the feasibility constraints of contract menus, in particular the incentive compatibility constraints for traders to report truthfully their preference type.

Lemma 2. *Under Assumption 1, and taking as given the cutoff type κ , a sufficient and necessary condition for the contract menus to be incentive compatible is that trading frequencies for buyers, q_a , are increasing in preference types and trading frequencies for sellers, q_b , are decreasing in preference types, and prices in the equilibrium contract menu follow the*

following equations,

$$q_{bx}p_{bx} = q_{bx}d_x + \int_x^\kappa q_{bs}d'_s ds, \quad \forall x \leq \kappa, \quad (17)$$

$$q_{ax}p_{ax} = q_{ax}d_x - \int_\kappa^x q_{as}d'_s ds, \quad \forall x \geq \kappa. \quad (18)$$

Lemma 2 helps simplify the problem of dealers: prices in the contract menu can be expressed as functions of trading frequencies. The proof of the lemma is a standard one used in the first order approach of mechanism design and is left in the Appendix. From equation (17) and (18), $p_{ax} < d_x$ for all $x \geq \kappa$, and $p_{bx} > d_x$ for all $x \leq \kappa$. So, the ex ante and ex post participation constraints are also satisfied. Lemma 2 gives the sufficient and necessary conditions for the feasibility of contract menus.

Following Lemma 1 and Lemma 2, dealers' problem can be simplified to the following. Let $D(n_{1\cdot}^m) \equiv D^a(n_{0\cdot}^m) + D^b(n_{1\cdot}^m)$, where $n_{0x}^m = \frac{\alpha}{\alpha+\gamma}g(x) - n_{1x}^m$. $D^a(n_{0\cdot}^m)$ is equal to the dealer's expected profit from selling assets to traders and $D^b(n_{0\cdot}^m)$ is equal to his expected profit from buying assets from traders. They are solved separately by equations (19) and (20).

$$\begin{aligned} rD^a(n_{0\cdot}^m) &= \max_{\{q_{ax}\}_{x \geq \kappa}} \int_\kappa^{x_H} \left[q_{ax}(p_{ax} - P)n_{0x}^m + f(x) \frac{\partial D^a(n_{0\cdot}^m)}{\partial n_{0x}^m} \right] dx \\ \text{s.t. } f(x) &= \alpha n_{0x}^u + \delta g(x)N_0^m - (\delta + \gamma + q_{ax})n_{0x}^m, \quad \forall x \geq \kappa, \\ q_{ax}p_{ax} &= q_{ax}d_x - \int_\kappa^x q_{as}d'_s ds, \quad \forall x \geq \kappa, \\ 0 \leq q_{ax} &\leq q, \quad q_{ax} \text{ is increasing in } x. \end{aligned} \quad (19)$$

$$\begin{aligned} rD^b(n_{1\cdot}^m) &= \max_{\{q_{bx}\}_{x \leq \kappa}} \int_{x_L}^{\kappa_t} \left[q_{bx}(P - p_{bx})n_{1x}^m + f(x) \frac{\partial D^b(n_{1\cdot}^m)}{\partial n_{1x}^m} \right] dx \\ \text{s.t. } f(x) &= \alpha n_{1x}^u + \delta g(x)N_1^m - (\delta + \gamma + q_{bx})n_{1x}^m, \quad \forall x \leq \kappa, \\ q_{bx}p_{bx} &= q_{bx}d_x + \int_x^\kappa q_{bs}d'_s ds, \quad \text{if } x \leq \kappa, \\ 0 \leq q_{bx} &\leq q, \quad q_{bx} \text{ is decreasing in } x. \end{aligned} \quad (20)$$

The first order conditions to these two problems can be characterized by ordinary differential equations, equations (34) and (35) in Section A.3 of the Appendix. I also show in the section that the solutions are optimal.

From equations (19) and (20), one can see the tradeoff facing the dealer when he decides the trading frequency. Take the ask contract of type $x > \kappa$ for example. Increasing the trading frequency with type x buyers increases the flow profit from trading with these buyers. But, it reduces the expected profit from all buyers of preference type above x , because it has a negative marginal effect on the expected payment, $q_{az}p_{az}$, for all $z > x$. This is the effect of IC constraints. Because the IC constraints are binding from above, increasing the trading frequency with type x increases the incentive for traders of type above x to deviate to the cheaper transaction price offered to type x , without losing as much immediacy. This is the information rent that the dealer needs to sacrifice when increasing the trading frequency. Since the payout of the information rent is larger for increasing the trading frequency with lower types, the dealer may want to stop trading altogether with traders with small gain from trade. Since the payout of the information rent is small when trading with traders with large gain, dealers increase the trading frequency to the upper bound.

A second negative effect of increasing the trading frequency comes from the *dynamic feedback*: increasing the trading frequency also reduces the continuation value for the dealer because it reduces the population density of those buyers in the future. Different from the first negative effect, the dynamic effect may be non-monotonic. The marginal dynamic effect of increasing the trading frequency with type x is $-\frac{\partial D(n_{0x}^m)}{\partial n_{0x}^m} n_{0x}^m$. $\frac{\partial D(n_{0x}^m)}{\partial n_{0x}^m} = \frac{q_{ax}(p_{ax}-P)}{r+\delta+\gamma+q_{ax}}$ is the expected profit of the dealer from trading with a buyer of type x . It depends on the trading frequency in the future and is an increasing function in q_{ax} in the future. n_{0x}^m is the density of buyers of type x . n_{0x}^m is proportional to $\frac{\alpha+\delta+\gamma}{\alpha+\delta+\gamma+\frac{\alpha+\delta}{\delta}q_{ax}}$. It depends on the trading frequency in the past and tends to be a decreasing function in q_{ax} in the past. The endogenous response of the density of buyers to the trading frequency is related to the Coase conjecture: with less buyers of higher types left because of trading in the past, the dealer increases the trading frequency with buyers of lower types, which induces the future self of the dealer to further increase the trading frequency with buyers of even lower types; in the limit where $\alpha, r, \delta, \gamma \ll q$, this will drive the transaction price all the way down the competitive market price, P . The conjecture partially breaks down here exactly because of

frictions in this environment. Since n_{0x}^m is proportional to $\frac{\alpha+\delta+\gamma}{\alpha+\delta+\gamma+\frac{\alpha+\delta}{\delta}q_{ax}}$, the matching friction, the frequency of preference shocks, stability of the relationship and the upper bound in the trading frequency all contribute to the partial breakdown.

Because of these two forces, the marginal dynamic effect may be non-monotonic in x and may be the largest in absolute value for traders with intermediate gain from trade. It is because of this dynamic effect that trade with traders with intermediate gain may be delayed in equilibrium.

Although the first order conditions for the dynamic optimization problem of the dealer are nonlinear equations in trading frequencies because of these marginal effects (see equations (34) and (35)), the dynamic optimization problem is actually linear. Because $\frac{\partial D(n_{0x}^m)}{\partial n_{0x}^m}$ is determined by the trading frequency in the future and n_{0x}^m is determined by the trading frequency in the past, they are taken as given in the optimization. The dealer actually solves a linear programming problem at any moment, since he takes as given trading frequencies in the future and the past, a consequence of the limited commitment of the dealer. Therefore, the local optimum is also optimal globally.

The following theorem shows that there always exists an equilibrium with non-trivial strategic delay, under a general regularity condition on the distribution function of preference types.

Theorem 1. *For a continuous type distribution $x \sim G(\cdot) \in C^1[x_L, x_H]$, there exists a stationary symmetric equilibrium where the trading frequencies in the equilibrium contract menus, $q_a : [x_L, x_H] \rightarrow [0, q]$ and $q_b : [x_L, x_H] \rightarrow [0, q]$, are continuous functions in preference type x .*

In the proof, I show that the equilibrium conditions can be formalized as solving for a fixed point problem for trading frequencies, q_a and q_b . I show that from the equilibrium conditions there exist a self mapping for the trading frequency function in the functional space of continuous functions in a closed space. Therefore, according to Brouwer's Fixed Point Theorem, the equilibrium exists. The proof is Section A.4 of the Appendix.

4 Implications on Market Liquidity

In this section, I will first present qualitative implications of the model on market liquidity. Then, I will show the effect of dispersion of valuation, frictions in forming and maintaining long-term trader-dealer relationships on market liquidity.

4.1 Qualitative Implications

Proposition 1. *Assume that the preference shock follows a continuous distribution. Then, the liquidity distortion is type specific in the stationary equilibrium with continuous trading frequencies. There exist threshold types $\underline{x}^b$, $\bar{x}^b$, κ , $\underline{x}^a$ and $\bar{x}^a$, with $x_L < \underline{x}^b < \bar{x}^b < \kappa < \underline{x}^a < \bar{x}^a < x_H$, such that*

- (1) *for sellers of type $x \in [\bar{x}^b, \kappa]$ and buyers of type $x \in [\kappa, \underline{x}^a]$, trade breaks down;*
- (2) *for sellers of type $x \in (\underline{x}^b, \bar{x}^b)$ and buyers of type $x \in (\underline{x}^a, \bar{x}^a)$, trade is delayed; the trading frequency is strictly lower than q and greater than 0, with the trading frequency for type x decreasing when x is closer to κ ; trade is carried out with strictly positive spread;*
- (3) *for sellers of type $x \in [x_L, \underline{x}^b]$ and buyers of type $x \in [\bar{x}^a, x_H]$, trade is carried out without delay; but the bid price for sellers of type $x \in [x_L, \underline{x}^b]$ is the lowest in the market and the ask price for buyers of type $x \in [\bar{x}^a, x_H]$ is the highest.*

See Section A.5 of the Appendix for the proof. The coexistence of the three regions categorized by the surplus from trade is a unique feature of the model. The literature on adverse selection with competitive search friction, initiated by Guerrieri, Shimer and Wright (2010)[16], only shows the second type of distortion, where all trade is delayed. In a static screening problem, where type distribution is not endogenous, trade either breaks down or is not delayed at all.

These features of equilibrium contract menus are illustrated in Figure 1. The parameter values used in the numerical example to generate the figure is in Table 1. In the numerical example, I set the annual discount rate at 0.05 following the numerical example in Duffie, Garleanu and Pedersen (2005)[7] (DGP). The arrival rate of the preference shock is set at

Preference			Matching		
shock arrival rate	δ	0.55	meeting rate	α	4
upper bound	x_H	11	breakup rate	γ	0.1
lower bound	x_L	9	Market		
discount rate	r	0.05	asset supply	A	0.5
distribution function	$G(x)$	$\frac{x-x_L}{x_H-x_L}$	trading frequency	q	365

Table 1: Parameter values used the numerical example.

0.55 to match the average rate in DGP. I assume that it takes on average three months for a trader to match with a dealer ($\alpha = 4$). And a trader-dealer relationship lasts on average 10 years ($\gamma = 0.1$). I assume that the trader and the dealer meet once every day. So, $q = 365$, which is 14 times as large as the trading frequency used in DGP. The dispersion of the preference shock affects the size of bid/ask spreads and information rent enjoyed by traders. This dispersion varies across assets, depending on such features as the opacity of the market and the sophistication of the asset under study. In this numerical example, I simply assume the preference is distributed uniformly on $[9, 11]$. The bid or ask spreads, defined as $\frac{|p_{ix}-P|}{P}$, $i \in \{b, a\}$, are presented in basis points in Figure 1. The spread for trading with probability 1 within a day is 3 basis points, which is around the estimated spread dealers charge for the highest trading volume in Edwards, Harris and Piwowar (2007)[9].

If traders have direct access to the interdealer market, it is easy to show that there is no trading delay in equilibrium and the equilibrium allocation is efficient and is the same as in an environment with complete information. The following Corollary summarizes the effect of the endogenous liquidity distortion on the aggregate allocation.

Corollary 1. *Compared to the equilibrium allocation in an environment where traders have direct access to the interdealer market, the equilibrium has the following qualitative properties: bid-ask spread arises; delay in trade arises; asset allocation is worse; aggregate trading flow is lower.*

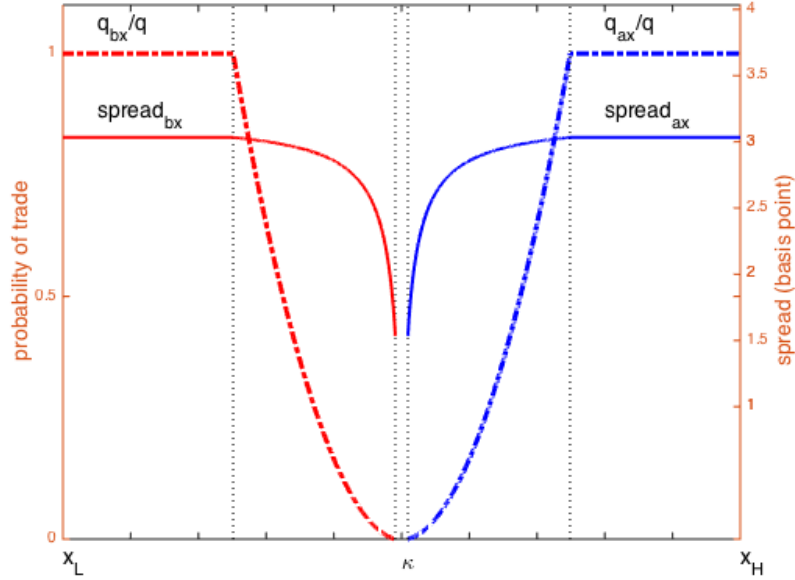


Figure 1: Equilibrium contract menus.

From Corollary 1, we can see that the equilibrium allocation of liquidity is not efficient. Figure 2 illustrates the inefficiency. The efficient allocation is calculated by assuming that all buyers and sellers trade at rate $q = 365$. The allocation in DGP is calculated by assuming that all trade takes place at rate $q = 26$, the value used in the numerical example in DGP.⁸ Quantitatively, the allocation in DGP is uniformly close to the efficient allocation for traders of all types. In contrast, allocation efficiency is type dependent in this model. The equilibrium allocation in this model is the same as the autarky allocation for traders with small gain, whose trade breaks down. For traders with intermediate gain, trade is delayed. The allocation for them is in between the efficient allocation and the autarky allocation. For traders with large gain, the allocation is the same as the efficient allocation, which is better than the DGP allocation. But quantitatively, the additional improvement appears small. This numerical exercise shows that quantitatively, the model could predict more severe asset

⁸To facilitate the comparison, what I call DGP in the exercise only allows trade between the dealer and those traders in long-term relationships with the dealer. Just that conditional on being matched together, they act as in DGP, trading at frequency $q = 26$.

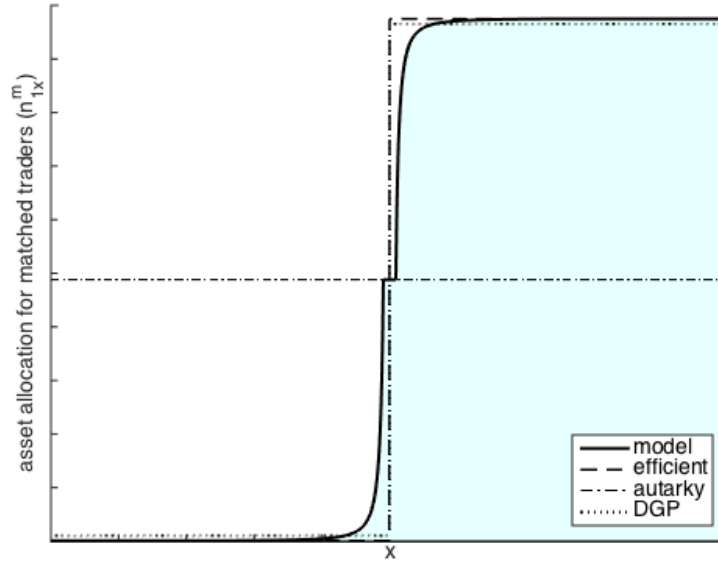


Figure 2: Equilibrium asset allocation.

misallocation. This is because the asset misallocation for traders with small or intermediate gain could be a lot more severe than that in DGP.

4.1.1 Empirical Implication on Liquidity Premium as a Decreasing Function of Trading Volume

Edwards, Harris and Piwowar (2007)[9] studies empirically how the opacity of the OTC markets of corporate bonds affects the transaction cost. They find that the liquidity premium, defined as the transaction cost per trading volume, is decreasing in trading volume. In other words, the total transaction cost is a concave function in trading volume. This relationship is not captured in benchmark models of the OTC market such as Lagos and Rocheteau (2005)[21], where the liquidity premium is an increasing function in trading volume.⁹ Lester,

⁹This is because of the concavity of the utility function. When the gain from trade for the trader is large, the marginal gain from trade is large and the trading volume is large. Therefore, the marginal gain from trade is increasing in trading volume and the liquidity premium, proportional to the average gain from trade, is also increasing in trading volume.

Rocheteau and Weill (2015)[23] shows that in an environment with directed search frictions, this relationship emerges because of the fixed cost to post contracts. Models with information frictions, such as Fuchs, Green and Papanikolaou (2014)[10], offer micro-foundations for the convex adjustment cost in reallocating assets. My paper falls into this category. It gives a partial answer to the link between opacity and the concave transaction cost in the OTC market.

Since the actual trading volume in my model is always 1 when a transaction is carried out, predictions in my model cannot be directly mapped to the empirical observation. But the contract menus can be thought of as a pricing schedule that charges bid or ask spread as a function of the expected trading volume¹⁰. Therefore, the model predicts that the bid/ask spread as a function of the expected trading volume has two components: a fixed component and a variable component. Regardless of the trading volume, dealers always charge a positive fixed spread. And the spread increases as the expected trading volume increases. If the expected trading volume is strongly correlated with the actual trading volume in the data, my model gives a micro-foundation to the empirical evidence on the negative relationship between the expected trading volume and the transaction cost.

Figure 3 compares the model prediction with the empirical finding. The point estimates of the liquidity premium (in basis points) from Edwards, Harris and Piwowar (2007)[9] are: 60 at 5k, 75 at 5k, 71 at 10k, 62 at 20k, 46 at 50k, 34 at 100k, 24 at 200k, 14 at 500k, 9 at 1000k, 5 at 2000k, 3 at 5000k and 4 at 10000k. I normalize the highest trading volume to 1. A caveat is that the point estimates are for the transaction cost per actual trading volume. I simply assume the expected trading volume in the data is the same as the actual trading volume. The model prediction uses parameter values in Table 1. Because the bid spread and ask spread are symmetric in this example, I use the bid spread to calculate the liquidity premium. The dispersion in preferences and the distribution function of the preference type in the table are not picked to match the empirical pattern. The good match between the

¹⁰The expected trading volume before a trader receives either a breakup shock or a preference shock is $\frac{q_{ix}}{q_{ix} + \delta + \gamma}$, where q_{ix} is the trading frequency.

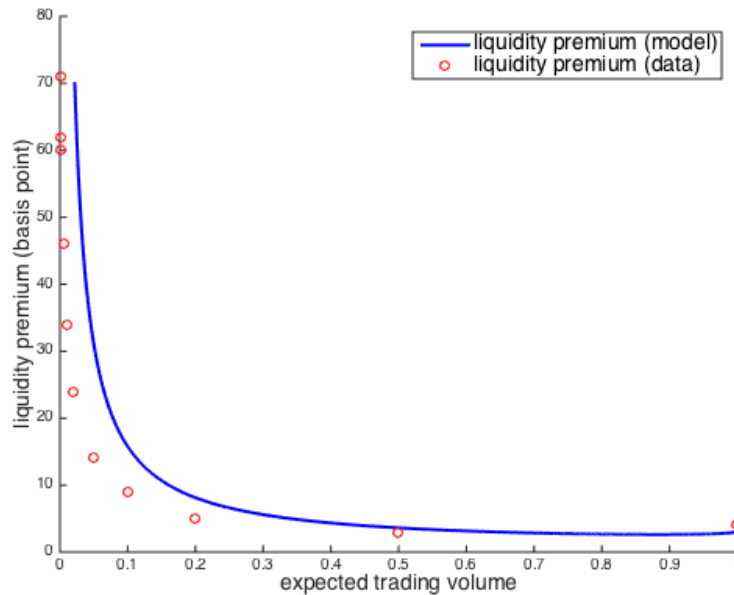


Figure 3: Liquidity premium and trading volume (expected trading volume for the model, actual trading volume for the data).

data and the model implies that the model could be a promising starting point to understand the liquidity distortion in the corporate bonds market.

Li and Schürhoff (2013)[24] shows that in the municipal bonds market, investors appear to face a trade-off between execution speed and cost. While the mechanism in my paper could contribute to the tradeoff, they also mentioned the role of inventories in providing immediacy. This is not captured in the model at all, because the interdealer market is assumed to be frictionless so that dealers do not need to hold inventories. To study the inventory risk associated with providing immediacy, one needs to have a better model about the interdealer trade.¹¹

¹¹See, for example, Chang and Zhang (2015)[5] for a recent attempt.

4.2 Comparative Statics

I will conduct comparative statics using the following two analytical examples. The solutions to the two examples are in Section A.7 and A.8 of the Appendix.

Example 1. Permanent Relationship

If we assume that dealer-trader relationships are permanent, the model can be solved in closed form. Assume that $\gamma = 0$, $A = \frac{1}{2}$ ¹² and that the type distribution is symmetric around $x_M = \frac{x_L + x_H}{2}$. Assume also that the distribution satisfies the following property: $g(x)(x - x_M)^2$ is increasing in x and $\frac{g(x)}{1-G(x)}$ is increasing in x when $x \geq x_M$. Then, $\kappa = x_M$,¹³

$$q_{ax} = \frac{g(x)}{g(\bar{x}^a)} \left(\frac{x - x_M}{\bar{x}^a - x_M} \right)^2 (\delta + q) - \delta, \forall x \in [\underline{x}^a, \bar{x}^a], \quad (21)$$

$$\bar{x}^a : 1 = \frac{g(\bar{x}^a) (\bar{x}^a - x_M)}{1 - G(\bar{x}^a)}, \quad (22)$$

$$\underline{x}^a : \left(\frac{\underline{x}^a - x_M}{\bar{x}^a - x_M} \right)^2 \frac{g(\underline{x}^a)}{g(\bar{x}^a)} = \frac{\delta}{\delta + q}, \quad (23)$$

and q_{bx} , $\bar{x}^b$, $\underline{x}^b$ can be derived in the same fashion.

Bunching could be optimal according to the first order conditions of the dealer's problem, equations (34) and (35). Under certain conditions for distribution functions of preference types, the trading frequencies are strictly monotonic when $q_x \in (0, q)$. In this example, I specified the condition on distributions to rule out bunching in equilibrium. The discount rate does not show up in the characterization not because it is zero but because dealers and traders share the same discount factor.

In this equilibrium, the qualitative properties in Proposition 1 and Corollary 1 hold. And we can also solve for bid or ask spreads.¹⁴ Qualitatively, the spread is higher when there is more delay. So, we focus on delay in comparative statics.

Example 2. Discrete Distribution

¹²The assumption on asset supply is just to simplify presentation.

¹³The characterization of bid contracts is symmetric and is omitted to economize the presentation.

¹⁴For example, if agents are patient in the sense that $r \ll 1$, $p_{ax} - P = \frac{\underline{x}^a - x_M}{\delta} + \int_{\underline{x}^a}^x \left[\frac{g(s)(s - x_M)^2}{g(s)(s - x_M)^2} - 1 \right] ds, \forall x \in [\underline{x}^a, \bar{x}^a]$.

Assume that the preference type follows a discrete distribution on $x_L < y < x_M < z < x_H$. $\Pr(x = x_L) = \Pr(x = x_H) = \pi_1$, $\Pr(x = y) = \Pr(x = z) = \pi_2$. $A = \frac{1}{2}$. $x_M = \frac{1}{2}(x_L + x_H)$, $x_M - y = z - x_M$. Further, assume that agents are very patient in the sense that $r \ll \delta$.

In this example, $q_{ax_H} = q_{bx_L} = q$, $q_{ax_M} = q_{bx_M} = 0$. Restricting our attention to parameter values such that $q_{az} = q_{by} \in (0, q)$, we have

$$q_{az} = \left[\varphi \frac{(\alpha + \delta) \left(1 + \frac{q}{\delta}\right) + \gamma}{\alpha + \delta + \gamma} - 1 \right] (\delta + \gamma). \quad (24)$$

where $\varphi = \frac{z - x_M}{x_H - z} \frac{\pi_2}{\pi_1}$. The participation constraints of intermediate types are binding in equilibrium. So, the spread for intermediate types does not depend on equilibrium delays and is equal to the surplus from trade. The spread for type x_L and x_H is decreasing in the delay facing intermediate types. This analytical example is particularly useful to analytically characterize the general equilibrium effect of the frictions in forming and maintaining long-term relationships.

4.2.1 Frictions Related to the Long-term Relationship

The general equilibrium effects of the frictions in forming and maintaining long-term relationships are summarized in Corollary 2 and Corollary 3. The two corollaries are derived from equation (24).

Corollary 2. *Under the conditions in Example 2, a smaller matching friction to form new relationships (higher α) corresponds to less equilibrium strategic delay in trade.*

The intuition behind Corollary 2 is the following. With a large matching friction (small α), it takes a long time for an unmatched trader to find a dealer. Then, the equilibrium asset allocation to unmatched traders is closer to the autarky allocation and is less affected by dealers' screening strategies, as is evident from the law of motion of traders, equation (11). Therefore, an unmatched trader is more likely to have large gain from trade when the matching friction is large. This gives dealers more incentive to screen those traders by delaying trade with other traders with smaller gain from trade. Therefore, the competition between

dealers is more intense with smaller matching friction. In terms of efficiency, Corollary 2 implies that the matching friction can be amplified by strategic delay of dealers.

Corollary 3. *Under the conditions in Example 2, the effect of relationship breakup on equilibrium strategic delay is ambiguous. If the matching friction is negligible ($\alpha \gg \max(\delta, \gamma)$), increasing the breakup rate decreases delay. If the matching friction is large ($\alpha \ll \delta$), increasing the breakup rate increases delay.*

The proof for the corollary is in Section A.9 of the Appendix. When the matching friction is small, relationship breakup induces more strategic interactions among dealers: screening by other dealers leaves few traders with large gain from trade in the population of unmatched traders; this in turn gives the dealer incentive to trade faster with those with intermediate gain from trade. But when the matching friction is large, the strategic interaction is muted. Meanwhile, a higher breakup rate means that the profit of a dealer depends more on the inflow of traders from the unmatched population, whose gain from trade is large because of the large matching friction. This induces dealers to delay trade with traders with intermediate gain from trade so as to screen traders with large gain from trade. Corollary 3 implies that the stability of the long-term relationship with dealers is more valuable for traders in an economy with fewer financial intermediaries, not only because of the exogenous frictions but also because it reduces strategic delay by dealers.

4.2.2 The Frequency of Preference Shocks

The effect of preference shocks on liquidity distortion is summarized in Corollary 4.

Corollary 4. *When the trader-dealer relationship is permanent, under the conditions of Example 1, equilibrium delay increases for all traders as the frequency of preference shocks increases. In particular, the measure of traders for whom trade breaks down increases ($\bar{x}^b$ decreases and $\underline{x}^a$ increases). Equilibrium delay in trade strictly increases for those traders facing delay in trade (buyers with preference type $x \in [\underline{x}^a, \bar{x}^a]$ and sellers with preference type $x \in [\underline{x}_b, \bar{x}^b]$).*

The proof is in Section A.10 of the Appendix. According to the law of motion of traders matched with dealers, equation (11), the asset allocation is closer to autarky allocation when the preference shock is more frequent. So, there are more traders with large gain from trade, this induces dealers to delay trade with traders with intermediate gain so as to screen traders with large gain from trade. Corollary 4 shows that in turbulent times, liquidity distortion could be higher because of strategic delay of the dealer.

Corollary 5. *Under the conditions in Example 2, increasing the frequency of preference shocks also induces more delay.*

Corollary 5 shows that findings in Corollary 4 is robust to the general equilibrium effects induced by the matching friction and relationship breakups. The proof in Section A.11 of the Appendix.

4.2.3 The Dispersion of Private Valuation

To use Example 1 to study the effect of valuation dispersion, assume that the distribution function satisfies the following functional form: $g(x) = \frac{1}{C} |x - x_M|^\zeta$ with $\zeta > -1$ and $C = \frac{(x_H - x_L)^{\zeta+1}}{2^{\zeta(\zeta+1)}}$. This class of distribution functions includes the uniform distribution as a special case ($\zeta = 0$). And a distribution function with a higher ζ has a thicker tail and higher dispersion. Corollary 6 shows that dispersed private valuation induces more delay.

Corollary 6. *Under the conditions in Example 2 and the conditions in Example 1 with $g(x) = \frac{1}{C} |x - x_M|^\zeta$ ($\zeta > -1$), increasing the dispersion of preference shocks increases trading delay for traders of all preference types.*

According to Corollary 6, then the dispersion increases, both the spread and delay in trade increase for traders of all preference types. But because there are more traders with large gain from trade in this case, the aggregate trading flow may not decrease.

Duffie, Malamud and Manso (2014)[8] shows that the opacity about heterogeneity in the asset quality also has important implications on trading activities in the market, and that the effect of opacity on price dispersion is not just driven by heterogeneity in asset quality but

also by the dispersion in private valuation. Edwards, Harris and Piwowar (2007)[9] studies empirically how the opacity of the OTC markets of corporate bonds affects the transaction cost. They find that the transaction cost is linked to the opacity of the OTC market. They get the result after controlling for factors related to heterogeneity in the asset quality, such as the credit risk. This implies that the private information about heterogeneous preferences could also be an important source of distortion when the market structure is opaque.

5 Other Implications

5.1 Implementation by Random Bid-Ask Quotes

In this section, I show that in an environment where dealers are restricted to trading without delay, the optimal contract menus can still be implemented. Imagine that a dealer hires a continuum of employees, each holding a telephone line. Each trader is assigned an employee to handle his trade. Employees are busy most of the time doing other things such as handling trade for other customers. So, the trader needs to wait for her turn, which arrives with Poisson rate q . When her turn comes, he gets a random draw from some distribution of bid/ask quotes decided by the dealer. And trade will be carried out without delay if the trader accepts the quote. Otherwise, the trader waits for his next turn to draw a quote from the distribution of bid/ask quotes specified by the dealer at that moment.

When dealers quote randomly, the strategy of traders can be characterized by their reservation prices, like in a random search model with price dispersion. Buyers will buy if and only if the ask price is below their reservation prices. Sellers will sell if and only if the bid price is above their reservation price. The reservation price is a function of traders' preference type. For example, buyers with a high preference type have a high reservation price. Therefore, they are more likely to accept an ask quote. Trade for those with large gain from trade is less likely to be delayed but the transaction cost is higher for them. This intuition is in line with the optimal contract menus designed by the dealer.

I show that if the dealer chooses optimally the distributions of bid and ask quotes, the

profit maximizing contract menus can be implemented. Denote the reservation price of traders with asset holding a and preference type x to be R_{ax} . Denote the distribution of ask quotes to be $F_a(\cdot)$ and the distribution of bid quotes to be $F_b(\cdot)$.

Lemma 3. *Given contract menus $\mathbb{M}_{ix} = \{(q_{ix}, p_{ix})\}_{\forall x \in \mathbb{X}, \forall i \in \{a, b\}}$, the price distributions $F_a(\cdot)$ and $F_b(\cdot)$ implementing the equilibrium contract menus satisfy the following equations,*

$$\frac{q_{ax}}{q} = F_a(R_{0x}), \quad \frac{q_{bx}}{q} = 1 - F_b(R_{1x}), \quad (25)$$

$$\frac{q_{ax}}{q} p_{ax} = \int_{R_{0\kappa}}^{R_{0x}} R dF_a(R), \quad \frac{q_{bx}}{q} p_{bx} = \int_{R_{1x}}^{R_{1\kappa}} R dF_b(R). \quad (26)$$

Lemma 3 gives a necessary condition for the optimal mechanism to be implementable by “random search”. But it is not sufficient because the reservation price solved this way must satisfy the indifference condition: $M_{1x} - R_{0x} = M_{0x}$ for buyers of type x and $M_{0x} + R_{1x} = M_{1x}$ for sellers of type x . The following lemma confirms these conditions.

Lemma 4. *Given the quote distributions implied by the optimal contract menus, R_{ax} is indeed the reservation price for traders with asset holding a and preference type x . $R_{ax} = d_x$.*

Lemma 3 and Lemma 4 together prove the following proposition.

Proposition 2. *The optimal mechanism can be implemented by dealers quoting randomly from price distributions $F_a(\cdot)$ and $F_b(\cdot)$ following equations (25) and (26).*

The price dispersion predicted by this implementation is greater than the price dispersion predicted by the contract menus. This is because the reservation price of a trader is equal to his valuation over an asset, d_x , and all traders whose trade does not break down enjoy some information rent, $p_{ax} < d_x$ for all $x \geq \underline{x}^a$, and $p_{bx} > d_x$ for all $x \leq \bar{x}^b$. We also know that the support of the distribution of ask(bid) prices is $[d_{\underline{x}^a}, d_{\bar{x}^a}]$ ($[d_{\underline{x}^b}, d_{\bar{x}^b}]$). But the difference in the price implication between the direct trading mechanism and this implementation is not quantitatively large. In the numerical example, the spread to trade with probability 1 in the direct trading mechanism (3.03 basis points) is close to the largest spread with this implementation (3.10 basis points).

The random pricing strategy of the dealer may also arise from competition among dealers. This mechanism is proposed by Burdett and Judd (1983)[2] and applied to study price dispersion in opaque financial markets by Green (2007)[14]. This implementation shows that even if the dealer has monopsony power, the price dispersion may still emerge in equilibrium as a result of dealers dynamically screening investors with heterogeneous valuations over the asset. Therefore, the paper offers an alternative mechanism linking the opacity of the secondary market to price dispersion.

5.2 Asset Pricing Implications

Proposition 3 shows that there exists an information premium in the asset price of the competitive interdealer market. And the premium depends on the *distribution of information rent* enjoyed by traders.

Proposition 3. *The asset price under asymmetric information has an additional term reflecting the information rent extracted by traders. The asset price satisfies the following equation,*

$$P = P_0 + \frac{c\mathcal{I}}{r}, \quad (27)$$

where

$$P_0 = \frac{\kappa}{r + \delta} + \frac{\delta Ex}{r(r + \delta)} \quad (28)$$

is the asset price under complete information, $Ex = \int x dG(x)$, c is a positive constant that depends on model parameters and $\mathcal{I}$ is a function of the information rent enjoyed by traders of different preference types.

$$\mathcal{I} = - \int_{\kappa}^{x_H} q_{ax}(d_x - p_{ax})dG(x) + \int_{x_L}^{\kappa} q_{bx}(p_{bx} - d_x)dG(x). \quad (29)$$

The detailed expression for c and $\mathcal{I}$ and the proof of the proposition are in Section A.14 of the Appendix.

$\mathcal{I}$ is nonzero only because of the information rent enjoyed by traders of different types. The ex post information rent is $p_{bx} - d_x$ for sellers of type x and $d_x - p_{ax}$ for buyers of type x .

$q_{ax}(d_x - p_{ax})$ and $q_{bx}(p_{bx} - d_x)$ represent the expected flow information rent. Under complete information, no surplus from trade is left to traders. So, $d_x = p_{ax}$ for all $x \geq \kappa$, $d_x = p_{bx}$ for all $x < \kappa$.

The expected information rent for buyers has a negative impact on the asset price while the expected information rent for sellers has a positive impact. So, depending on the distribution of the information rent, the information premium may be either positive or negative. The following example highlights the effect of the preference distribution on the information premium. Given any continuous distribution function, $G(x)$, consider the following transformations such that either the trade with buyers or sellers generates no surplus,

$$\begin{aligned} G^-(x) &= \begin{cases} G(x), & \text{if } x < \kappa_0, \\ 1, & \text{if } x \geq \kappa_0, \end{cases} \\ G^+(x) &= \begin{cases} 0, & \text{if } x < \kappa_0, \\ G(x), & \text{if } x \geq \kappa_0, \end{cases} \end{aligned}$$

where κ_0 is such that $G(\kappa) = \frac{1}{2}$. Suppose the distribution function $G(x)$ has zero skewness. Then the skewness of $G^-(x)$ is negative. The skewness of $G^+(x)$ is positive.

Corollary 7. *Given any continuous distribution function $G(\cdot)$ on interval $[x_L, x_H]$, and asset supply $A = \frac{1}{2}$, the information premium, $\frac{c\mathcal{I}}{r}$, is positive for preference distribution $G^-(\cdot)$ and negative for preference distribution $G^+(x)$.*

In both cases in the corollary, the preference type of the marginal trader is κ_0 . So the asset prices varies only with the information premium. The information premium is positive with the negatively skewed distribution, $G^-(\cdot)$ and negative with the positively skewed distribution, $G^+(\cdot)$.

The distribution of information rent also depends on the total asset supply because the asset supply affects the information rent distributed to buyers and sellers. A shift in the supply not only shifts the preference type of the marginal trader but also shifts the conditional distribution of preference types of buyers and sellers. In general, the strategic delay of dealers affects asset prices not only through the information premium but also through the preference type of the marginal trader, κ .

The effect of the information friction on asset prices also depends on other frictions in the market. For example, the constant c in asset pricing equation (27) depends on the matching friction, the breakup rate of the trader-dealer relationship and the arrival rate of preference shocks.

Since the information premium is related to the strategic delay of the dealer, empirically it can be part of the liquidity premium in the asset price. Assets typically associated with positive liquidity premium, such as the short-run or on-the-run treasury securities, are typically used to hedge against negative preference shocks. Consistent with the observation that these securities have positive liquidity premium, the information premium is positive in this case. The opacity in such OTC markets as the corporate bonds market may affect the asset price through the information premium.

6 Conclusion

I study in an OTC market the effect of unobservable private valuation on market liquidity and asset allocation when dealers and traders are in long-term relationships. I solve the optimal trading mechanism contingent on the aggregate history of the traders matched with a dealer, summarized by the asset allocation. The dynamic screening behavior of the dealer provides an additional mechanism to account for the liquidity distortion, price dispersion and the asset price in the market.

A Appendix

From equations (5), (6), (7) and (8), we can show that

$$(r + q_{axt} + q_{bxt} + \gamma + \delta)d_{xt} = x + q_{axt}p_{axt} + q_{bxt}p_{bxt} + \gamma e_{xt} + \delta E d_t + \dot{d}_{xt}, \quad (30)$$

$$(r + \gamma + \delta)e_{xt} = x + \alpha d_{xt} + \delta E e_t + \dot{e}_{xt}, \quad (31)$$

where $Ee_t = \int e_{xt}dG(x)$, $Ed_t = \int d_{xt}dG(x)$. Using the Envelope Theorem, these two functions are differentiable and the derivatives are characterized by the following equations,

$$(r + q_{axt} + q_{bxt} + \gamma + \delta)d'_{xt} = 1 + \gamma e'_{xt} + \dot{d}'_{xt}, \quad (32)$$

$$(r + \gamma + \delta)e'_{xt} = 1 + \alpha d'_{xt} + \dot{e}'_{xt}. \quad (33)$$

Notice that if q_{axt} and q_{bxt} is not continuous, d_{xt} and e_{xt} may not be differentiable everywhere. The continuity of q_{axt} and q_{bxt} will be proved in the Appendix.

A.1 Proof for Lemma 1

By the ex post IR constraints of the trader, $p_{bx} \geq M_{1x} - M_{0x}$ for all type x willing to sell. $M_{1x} - M_{0x} \geq p_{ax}$ for all type x willing to buy. Since $p_{bx} \leq P \leq p_{ax}$, the proof is finished if we can show there exists a unique type κ such that $d_\kappa = P$. This is true if d_x is strictly increasing in x .

Since q_a and q_b are continuous almost everywhere. So, we can apply the Envelope Theorem almost everywhere to derive the derivative of the value functions. For those preference types where both q_a and q_b are continuous, M'_{ix} and U'_{ix} can be written as follows,

$$rM'_{1x} = 1 - \delta M'_{1x} - q_{bx}d'_x + \gamma(U'_{1x} - M'_{1x})$$

$$rU'_{1x} = 1 + \alpha(M'_{1x} - U'_{1x}) - \delta U'_{1x}$$

$$rM'_{0x} = -\delta M'_{0x} + q_{ax}d'_x + \gamma(U'_{0x} - M'_{0x})$$

$$rU'_{0x} = \alpha(M'_{0x} - U'_{0x}) - \delta U'_{0x}$$

So d'_x and e'_x are as follows,

$$(r + q_{ax} + q_{bx} + \gamma + \delta)d'_x = 1 + \gamma e'_x,$$

$$(r + \gamma + \delta)e'_x = 1 + \alpha d'_x,$$

from which we can derive $d'_x = \frac{1 + \gamma \frac{1}{r + \alpha + \delta}}{r + \delta + q_{bx} + q_{ax} + \gamma \frac{r + \delta}{r + \alpha + \delta}} > 0$. Therefore, d_x is differentiable almost everywhere and it is a continuous and strictly increasing function. Q.E.D.

A.2 Proof for Lemma 2

I only present the proof for the conditions for ask contracts here. The proof for the conditions for bid contracts is completely symmetric and is thus omitted.

First, from Lemma 1, d_x is strictly increasing in x and is differentiable almost everywhere.

Necessity. we can show that given any two contracts, (q_{am}, p_{am}) and $(q_{a'm'}, p_{a'm'})$ with $q \geq q_{am} > q_{a'm'} \geq 0$, for any $x_H \geq x > x' \geq \kappa$, if $q_{am}(d_{x'} - p_{am}) \geq q_{a'm'}(d_{x'} - p_{a'm'})$, then $q_{am}(d_x - p_{am}) \geq q_{a'm'}(d_x - p_{a'm'})$. This is because

$$\begin{aligned} q_{am}(d_x - p_{am}) &= q_{am}(d_x - d_{x'}) + q_{am}(d_{x'} - p_{am}) \\ &\geq q_{am}(d_x - d_{x'}) + q_{a'm'}(d_{x'} - p_{a'm'}) \\ &> q_{a'm'}(d_x - d_{x'}) + q_{a'm'}(d_{x'} - p_{a'm'}) \\ &= q_{a'm'}(d_x - p_{a'm'}). \end{aligned}$$

This implies that if we index the contract in menu $\mathbb{M}_a$ by the preference type of traders picking the contract, then q_{ax} must be weakly increasing in x .

Since $x \in \arg \max_m q_{am}(d_x - p_{am})$ and from Lemma 1, d_x is differentiable almost everywhere, applying the Envelope Theorem where d_x is differentiable, we have $q_{ax}(d_x - p_{ax}) = \int_{\kappa}^x q_{as} d'_s ds$, which means that $q_{ax} p_{ax} = q_{ax} d_x - \int_{\kappa}^x q_{as} d'_s ds$.

Sufficiency. It is optimal for a trader of type x to choose the contract with index x . Comparing her payoff from choosing contract x and contract m , we have,

$$\begin{aligned} & q_{ax}(d_x - p_{ax}) - q_{am}(d_x - p_{am}) \\ &= \int_{\kappa}^x q_{as} d'_s ds - \int_{\kappa}^m q_{as} d'_s ds - q_{am}(d_x - d_m) \\ &= \int_m^x (q_{as} - q_{am}) d'_s ds \geq 0, \forall m \in [x_L, x_H]. \end{aligned}$$

The first equality uses the condition that $q_{ax} p_{ax} = q_{ax} d_x - \int_{\kappa}^x q_{as} d'_s ds$. The inequality uses the condition that q_{am} is weakly increasing in m . Q.E.D.

A.3 Dynamic Optimization of Dealers

The HJB equation for dealers is

$$\begin{aligned}
rD_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) &= \max \int \left[q_{axt}(p_{axt} - P_t)\tilde{n}_{0xt}^m + g_{0x} \frac{\partial}{\partial \tilde{n}_{0xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) \right. \\
&\quad \left. + q_{bxt}(P_t - p_{bxt})\tilde{n}_{1xt}^m + g_{1x} \frac{\partial}{\partial \tilde{n}_{1xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) \right] dx \\
s.t. \quad \frac{d}{dt} \tilde{n}_{1xt}^m &= g_{1x} = \alpha n_{1xt}^u + \delta g(x) \tilde{N}_1^m - (\delta + \gamma + q_{bxt}) \tilde{n}_{1xt}^m, \forall x < \kappa_t, \\
\frac{d}{dt} \tilde{n}_{0xt}^m &= g_{0x} = \alpha n_{0xt}^u + \delta g(x) \tilde{N}_0^m - (\delta + \gamma + q_{axt}) \tilde{n}_{0xt}^m, \forall x > \kappa_t, \\
q_{bxt} p_{bxt} &= q_{bxt} d_{xt} + \int_x^\kappa q_{bst} d'_{st} ds, \text{ if } x \leq \kappa_t, \\
q_{axt} p_{axt} &= q_{axt} d_{xt} - \int_\kappa^x q_{ast} d'_{st} ds, \text{ if } x > \kappa_t, \\
0 \leq q_{bx} &\leq q, \quad q_{bx} \text{ is decreasing in } x. \\
0 \leq q_{ax} &\leq q, \quad q_{ax} \text{ is increasing in } x.
\end{aligned}$$

Or,

$$\begin{aligned}
rD_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) &= \max_{\{q_{axt}, q_{bxt}\}} \int_{\kappa_t}^{x_H} \left[(q_{axt} d_{xt} - \int_\kappa^x q_{ast} d'_{st} ds - q_{axt} P_t) \tilde{n}_{0xt}^m + g_{0x} \frac{\partial}{\partial \tilde{n}_{0xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) \right. \\
&\quad \left. + \gamma_{1ax}(q - q_{axt}) + \gamma_{0ax} q_{axt} + \eta_{ax} \left(q_{axt} - \max_{\kappa_t \leq s < x} q_{ast} \right) \right] dx \\
&\quad + \int_{x_L}^{\kappa_t} \left[(q_{bxt} P_t - q_{bxt} d_{xt} - \int_x^\kappa q_{bst} d'_{st} ds) \tilde{n}_{1xt}^m + g_{1x} \frac{\partial}{\partial \tilde{n}_{1xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) \right. \\
&\quad \left. + \gamma_{1bx}(q - q_{bxt}) + \gamma_{0bx} q_{bxt} + \eta_{bx} \left(q_{bxt} - \max_{x < s \leq \kappa_t} q_{bst} \right) \right] dx \\
&\quad + \frac{d}{dt} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m),
\end{aligned}$$

where γ_{1ix} is the multiplier for inequality $q \geq q_{ixt}$, γ_{0ix} is the multiplier for inequality $q_{ixt} \geq 0$ and η_{ix} is the multiplier for the monotonicity condition, $q_{ixt} \geq \max_{\kappa_t \leq s < x} q_{ist}$ if $i = a$, $q_{ixt} \geq \max_{x < s \leq \kappa_t} q_{ist}$ if $i = b$. η_{ix} is positive if there is bunching in the solution and type x falls in the bunching region. For $x \geq \kappa_t$, the first order condition with respect to q_{axt} is

$$\begin{aligned}
0 &= (d_{xt} - P_t) \tilde{n}_{0xt}^m - d'_{xt} \int_x^{x_H} \tilde{n}_{0st}^m ds - \frac{\partial D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m)}{\partial \tilde{n}_{0xt}^m} \tilde{n}_{0xt}^m \\
&\quad - \gamma_{1ax} + \gamma_{0ax} + \eta_{ax} - \int \eta_{as} \mathbb{I}_{\{q_{as} = q_{as}\}} ds
\end{aligned}$$

The second derivative of the objective function with respect to q_{axt} is 0.

Likewise, for $x \leq \kappa_t$, the first order condition with respect to q_{bxt} is

$$\begin{aligned} 0 = & (P_t - d_{xt})\tilde{n}_{1xt}^m - d'_{xt} \int_{x_L}^x \tilde{n}_{1st}^m ds - \frac{\partial}{\partial \tilde{n}_{1xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) \tilde{n}_{1xt}^m \\ & - \gamma_{1bx} + \gamma_{0bx} - \eta_{bx} + \int \eta_{bs} \mathbb{I}_{\{q_{bx}=q_{bs}\}} ds. \end{aligned}$$

And the second derivative of the objective function with respect to q_{bxt} is also 0.

Because this is a linear problem, locally optimal solutions are also globally optimal.

At the steady state, we can conjecture and verify that $D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m)$ is a linear function in $\tilde{n}_{0xt}^m$ and $\tilde{n}_{1xt}^m$, $\frac{\partial^2}{\partial \tilde{n}_{0xt}^m \partial \tilde{n}_{0xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) = \frac{\partial^2}{\partial \tilde{n}_{0xt}^m \partial \tilde{n}_{1xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) = \frac{\partial^2}{\partial \tilde{n}_{1xt}^m \partial \tilde{n}_{1xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) = 0$.

For $x \geq \kappa$,

$$\begin{aligned} r \frac{\partial}{\partial \tilde{n}_{0xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) &= (q_{axt} d_{xt} - \int_{\kappa}^x q_{ast} d'_{st} ds - q_{axt} P_t) - (\delta + \gamma + q_{axt}) \frac{\partial}{\partial \tilde{n}_{0xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m), \\ \frac{\partial}{\partial \tilde{n}_{0xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) &= \frac{q_{axt} d_{xt} - \int_{\kappa}^x q_{ast} d'_{st} ds - q_{axt} P_t}{r + \delta + \gamma + q_{axt}}. \end{aligned}$$

Likewise, for $x < \kappa$, we can show that

$$\frac{\partial}{\partial \tilde{n}_{1xt}^m} D_t(\tilde{n}_{0xt}^m, \tilde{n}_{1xt}^m) = \frac{q_{bxt} P_t - q_{bxt} d_{xt} - \int_x^{\kappa} q_{bst} d'_{st} ds}{r + \delta + \gamma + q_{bxt}}.$$

Given κ , the equation that solve for the menu of ask contracts is,

$$\begin{aligned} -\frac{d}{dx} \left[\left(\int_{\kappa}^x \frac{r + \delta + \gamma + q_{as}}{r + \delta + \gamma + q_{ax}} d'_s ds \right) \left(\int_x^{x_H} n_{0s}^m ds \right) \right] \\ - \gamma_{1ax} + \gamma_{0ax} + \eta_{ax} - \int \eta_{as} \mathbb{I}_{\{q_{ax}=q_{as}\}} ds = 0, \end{aligned} \quad (34)$$

where

$$\begin{aligned} d'_x &= \frac{r + \alpha + \delta + \gamma}{(r + \alpha + \delta)(r + \gamma + \delta + q_{ax}) - \alpha\gamma}, \\ n_{0x}^m &= g(x) \frac{\alpha}{\alpha + \gamma} \frac{\alpha + \delta + \gamma}{\alpha + \delta + \gamma + \frac{\alpha + \delta}{\delta} q_{ax}} (1 - A). \end{aligned}$$

Likewise, the equation that solve for the menu of bid contracts is,

$$\begin{aligned} -\frac{d}{dx} \left[\left(\int_x^{\kappa} \frac{r + \delta + \gamma + q_{bs}}{r + \delta + \gamma + q_{bx}} d'_s ds \right) \left(\int_{x_L}^x n_{1s}^m ds \right) \right] \\ - \gamma_{1bx} + \gamma_{0bx} - \eta_{bx} + \int \eta_{bs} \mathbb{I}_{\{q_{bx}=q_{bs}\}} ds = 0, \end{aligned} \quad (35)$$

where

$$\begin{aligned} d'_x &= \frac{r + \alpha + \delta + \gamma}{(r + \alpha + \delta)(r + \gamma + \delta + q_{bx}) - \alpha\gamma}, \\ n_{1x}^m &= g(x) \frac{\alpha}{\alpha + \gamma} \frac{\alpha + \delta + \gamma}{\alpha + \delta + \gamma + \frac{\alpha + \delta}{\delta} q_{bx}} A. \end{aligned}$$

In the derivation, we take as given q_{ax} and q_{bx} are continuous functions. This is a conjecture. It will be verified in Section A.4 of the Appendix.

A.4 Proof of the Existence Theorem, Theorem 1

First, I prove there exists a solution for the optimal bid and ask contract menu, given κ corresponding to the market clearing price ($d_\kappa = P$). Then, I will show that given the contract menus as functions of the cutoff type κ , there exists a cutoff type κ such that the interdealer market clears.

To show that there exists a solution for the optimal ask contract menu, I construct a mapping from a continuous and increasing function $q_a : [\kappa, x_H] \rightarrow [0, q]$ to another continuous and increasing function $\hat{q}_a : [\kappa, x_H] \rightarrow [0, q]$ using the first order condition from dealers' optimal control problem.¹⁵ For all $x \in [\kappa, x_H]$,

$$\hat{q}_{ax} = \begin{cases} 0, & \text{if } \tilde{q}_{ax} \leq 0, q_{as} \leq 0, \forall s < x, \\ q, & \text{if } \tilde{q}_{ax} \geq q, \\ \max_{s \leq x} \{ \min(q, \max(0, \tilde{q}_{as})) \}, & \text{otherwise,} \end{cases}$$

where

$$\tilde{q}_{ax} = \frac{r + \alpha + \delta + \gamma}{r + \alpha + \delta} \left[\frac{1}{d_{1x}} - (r + \delta) \right],$$

$d_{1x} = n_{0x}^m \int_\kappa^x \frac{r + \delta + \gamma + q_{as}}{r + \delta + \gamma + q_{ax}} d'_s ds \left(\int_x^{x_H} n_{0s}^m ds \right)^{-1}$ and $n_{0x}^m = g(x) \frac{\alpha}{\alpha + \gamma} \frac{\alpha + \delta + \gamma}{\alpha + \delta + \gamma + (\alpha + \delta) q_{ax} / \delta} (1 - A)$. Therefore, according to the Brouwer Fixed Point Theorem, there exists a fixed point for the

¹⁵For the fixed point argument to work, the monotonicity is not essential. But the monotonicity holds anyway because of Lemma 2.

mapping from q_a to $\hat{q}_a$. At the fixed point, the first order condition from dealers' optimal control problem is satisfied. With the same procedure, there exists an equilibrium bid contract menu given κ .

Given the contract menus as functions of the cutoff type κ , the demand for assets must be no greater than the supply if $\kappa = x_H$, and the demand must be no less than the supply if $\kappa = x_L$. There must exist a κ that clears the market if the demand and the supply changes continuously in κ . From the first order conditions of dealers' optimal control problem, which the optimal contract menus satisfy, we can show using Implicit Function Theorem that q_{ax} , for all $x \geq \kappa$, and q_{bx} , for all $x \leq \kappa$, are differentiable with respect to κ . Then, the demand and the supply are continuous in κ .

So far, we showed that there exists an equilibrium where q_a is continuous for all $x \geq \kappa$, q_b is continuous for all $x \leq \kappa$. What remains to be shown is there there is q_a and q_b is continuous at $x = \kappa$. Here, we can borrow the arguments in the proof for Proposition 1, in Section A.5 of the Appendix. Q.E.D.

A.5 Proof of Proposition 1

I will present only the proof for the ask contract menu. The proof for the bid contract menu is symmetric. q_a is characterized by equation (34), which allows for bunching and hitting the lower bound of the trading frequency, 0, and the upper bound of the trading frequency, q . A key to the characterization is the term, $F(x) = \left(\int_{\kappa}^x \frac{r+\delta+\gamma+q_{as}}{r+\delta+\gamma+q_{ax}} d'_s ds \right) \left(\int_x^{x_H} n_{0s}^m ds \right)$, in the equation, which is a differentiable and nonnegative function.

$$\begin{aligned} F'(\kappa) &= d'_{\kappa} \int_{\kappa}^{x_H} n_{0s}^m ds > 0, \\ F'(x_H) &= -n_{0x_H}^m \int_{\kappa}^{x_H} \frac{r+\delta+\gamma+q_{as}}{r+\delta+\gamma+q_{ax}} d'_s ds < 0. \end{aligned}$$

At $x = \kappa$, $q_{a\kappa} = 0$. Since, $q_{ax} \geq 0$, for all $x \geq 0$, $\eta_{a\kappa} - \int \eta_{as} \mathbb{I}_{\{q_{a\kappa}=q_{as}\}} ds = 0$. So equation (34) is simplified as $-\frac{d}{dx} F(\kappa) + \gamma_{0a\kappa} = 0$. According to Theorem 1, q_a in equilibrium is a continuous

function on the interval $(\kappa, x_H]$.¹⁶ Therefore, $\frac{d}{dx}F(x)$ is continuous on the interval $[\kappa, x_H]$. This means that there exists an $x_{La} > \kappa$, such that which $\frac{d}{dx}F(x) > 0$ for all $x \in [\kappa, x_{La}]$. Therefore, $\gamma_{0ax} = \frac{d}{dx}F(x) > 0$, $q_{ax} = 0$ for all $x \in [\kappa, x_{La}]$. Similarly, I can show that there exists an $x_{Ha} < x_H$ such that $q_{ax} = q$ for all $x \in [x_{Ha}, x_H]$.

According to Lemma 2, q_a is increasing in x . Therefore, define, $\underline{x}^a = \sup\{x : q_{ax} = 0\}$ and $\bar{x}^a = \inf\{x : q_{ax} = q\}$. We have $\underline{x}^a \geq x_{La} > \kappa$, and $\bar{x}^a \leq x_{Ha} < x_H$. And for all $x \in [\kappa, \underline{x}^a]$, $q_{ax} = 0$. For all $x \in [\bar{x}^a, x_H]$, $q_{ax} = q$. Since q_a is a continuous function, $\underline{x}^a < \bar{x}^a$. Because of the definition of $\bar{x}^a$ and $\underline{x}^a$ and the fact that q_{ax} is increasing in x , we have $0 < q_{ax} < q$, for all $x \in (\underline{x}^a, \bar{x}^a)$. Q.E.D.

A.6 Proof of Corollary 1

From Theorem 1, we can show that $\bar{x}^b < \kappa < \underline{x}^a$. This means that dealers must earn positive profits. As long as dealers earn positive profits, there must exist non-trivial bid-ask spread. No trade takes place for traders with preference type $x \in [\bar{x}^b, \underline{x}^a]$. So, there are delays due to endogenous trading frictions and trading volume must decrease.

Under efficient allocation, there is no delay in trade for any trader. Denote κ^* to be the threshold type for the efficient allocation. If $\kappa^* \geq \kappa$, the flow of asset supply from sellers is smaller than the trading flow under complete information, because there exists trading delay for some seller types. Likewise, if $\kappa^* < \kappa$, the flow of asset demand from buyers is smaller than the trading flow under complete information. Therefore, the trading flow with asymmetric information must be lower than that with efficient allocation. Q.E.D.

A.7 Analytical Example 1 ($\gamma = 0$)

I present the solution for ask contracts here. The solution for bid contracts follows the same procedure. For $x \in (\underline{x}^a, \bar{x}^a)$, the Lagrangian multipliers $\gamma_{1ax} = \gamma_{0ax} = 0$. Suppose the solution for q_{ax} is strictly increasing in this region, $\eta_{ax} = 0$. This conjecture can be verified

¹⁶We only need continuity of the function on $(\kappa, x_H]$. Whether q_{ax} is continuous at $x = \kappa$ relies on the rest of the proof of the proposition.

later. Then, according to equation (34),

$$\frac{\int_{\kappa}^x (r + \delta + q_{as}) d'_s ds}{(r + \delta + q_{ax}) d'_x} = \frac{\int_x^{x_H} \tilde{n}_{0s}^m ds}{\tilde{n}_{0x}^m}, \forall x \in (\underline{x}^a, \bar{x}^a), \quad (36)$$

where

$$\begin{aligned} d'_x &= \frac{1}{r + \delta + q_{ax}}, \\ \tilde{n}_{0x}^m &= g(x) \frac{\delta}{\delta + q_{ax}} (1 - A). \end{aligned}$$

Let $F(x) = \int_x^{x_H} \frac{g(s)}{\delta + q_{as}} ds$. The ordinary differential equation (36) can be simplified to be

$$\begin{aligned} \frac{d}{dx} \ln F(x) &= -\frac{1}{x - \kappa}, \forall x \in (\underline{x}^a, \bar{x}^a), \\ F(\bar{x}^a) &= \frac{1 - G(\bar{x}^a)}{\delta + q}. \end{aligned}$$

The boundary condition at $x = \bar{x}^a$ uses the fact that $q_{ax} = q$, for all $x > \bar{x}^a$. From the simplified ODE, we can solve for q_{ax} . $\forall x \in (\underline{x}^a, \bar{x}^a)$,

$$q_{ax} = g(x) \frac{(x - \kappa)^2}{\bar{x}^a - \kappa} \frac{\delta + q}{1 - G(\bar{x}^a)} - \delta. \quad (37)$$

At $\bar{x}^a$, $q_{a\bar{x}^a} = q$. So, from equation (37), $\bar{x}^a$ satisfies the following equation: $1 = \frac{g(\bar{x}^a)(\bar{x}^a - \kappa)}{1 - G(\bar{x}^a)}$.

At $\underline{x}^a$, $q_{a\underline{x}^a} = 0$. From equation (37), $\underline{x}^a$ satisfies the following equation: $\frac{\delta}{\delta + q} = \left(\frac{\underline{x}^a - \kappa}{\bar{x}^a - \kappa} \right)^2 \frac{g(\underline{x}^a)}{g(\bar{x}^a)}$.

The spread $p_{ax} - P$ satisfies the following equation,

$$\begin{aligned} q_{ax}(p_{ax} - P) &= \int_{\kappa}^x (q_{ax} - q_{as}) d'_s ds \\ &= \int_{\underline{x}^a}^x \frac{g(x)(x - \kappa)^2 - g(s)(s - \kappa)^2}{g(\bar{x}^a)(\bar{x}^a - \kappa)^2} \frac{\delta + q}{r + \frac{g(s)}{g(\bar{x}^a)} \left(\frac{s - \kappa}{\bar{x}^a - \kappa} \right)^2 (\delta + q)} ds + q_{ax} \frac{(\underline{x}^a - \kappa)}{r + \delta}, \end{aligned}$$

which implies that

$$p_{ax} - P = \frac{\underline{x}^a - \kappa}{r + \delta} + \frac{1}{q_{ax}} \int_{\underline{x}^a}^x \frac{g(x)(x - \kappa)^2 - g(s)(s - \kappa)^2}{g(\bar{x}^a)(\bar{x}^a - \kappa)^2} \frac{\delta + q}{r + \frac{g(s)}{g(\bar{x}^a)} \left(\frac{s - \kappa}{\bar{x}^a - \kappa} \right)^2 (\delta + q)} ds.$$

If $r \ll 1$,

$$p_{ax} - P = \frac{\underline{x}^a - \kappa}{\delta} + \int_{\underline{x}^a}^x \left[\frac{g(x)(x - \kappa)^2}{g(s)(s - \kappa)^2} - 1 \right] ds.$$

A.8 Analytical Example 2 (Discrete Distribution)

Assume that the preference type follows a discrete distribution on $x_L < y < x_M < z < x_H$. $\Pr(x = x_L) = \Pr(x = x_H) = \pi_1$, $\Pr(x = y) = \Pr(x = z) = \pi_2$. $A = \frac{1}{2}$. $x_M = \frac{1}{2}(x_L + x_H)$, $x_M - y = z - x_M$. Because the distribution is symmetric, I will focus on solving the ask contracts. Because of symmetry, $\kappa = x_M = P$. Dealers earn no profit selling assets to traders of type x_M . So, $q_{akt} = 0$. And there is no point to delay the trade with traders of type x_H . So, $q_{ax_Ht} = q$. Because type z traders are the lowest type dealers are going to sell assets to, $p_{azt} = d_{zt}$. And since it is optimal for dealers to make the IC constraint of type x_H traders bind, $q(d_{x_Ht} - p_{x_Ht}) = q_{azt}(d_{x_Ht} - d_{zt})$. Given these conditions, the ask contract menu is solved by the following problem,

$$\begin{aligned} rD_t^a(n_{0zt}^m) &= \max_{q_{azt}, p_{x_Ht}} q_{azt}(d_{zt} - P_t)n_{0zt}^m + q(p_{x_Ht} - P_t)n_{0x_H}^m + g_{0z} \frac{\partial D_t^a(n_{0zt}^m)}{\partial n_{0zt}^m}, \\ \text{s.t.} \quad &q(d_{x_Ht} - p_{x_Ht}) = q_{azt}(d_{x_Ht} - d_{zt}), \\ &g_{0z} = \alpha n_{0z}^u + \delta \pi_2 N_0^m - (\delta + \gamma + q_{azt})n_{0zt}^m, \\ &0 \leq q_{azt} \leq q, \end{aligned}$$

where $n_{0x_H}^m = \pi_1 \frac{\alpha}{\alpha + \gamma} \frac{\alpha + \delta + \gamma}{\alpha + \delta + \gamma + \frac{\alpha + \delta}{\delta} q} (1 - A)$. The first order condition of the problem at the steady state is

$$(d_z - P)n_{0z}^m \frac{r + \delta + \gamma}{r + \delta + \gamma + q_{az}} - (d_{x_H} - d_z)n_{0x_H}^m - \gamma_{1az} + \gamma_{0az} = 0,$$

where $n_{0z}^m = \pi_2 \frac{\alpha}{\alpha + \gamma} \frac{\alpha + \delta + \gamma}{\alpha + \delta + \gamma + \frac{\alpha + \delta}{\delta} q_{az}} (1 - A)$. Assume that $r \ll \delta$.

$$q_{az} = \left[\frac{z - x_M}{x_H - z} \frac{\pi_2}{\pi_1} \frac{(\alpha + \delta) \left(1 + \frac{q}{\delta}\right) + \gamma}{\alpha + \delta + \gamma} - 1 \right] (\delta + \gamma).$$

A.9 Proof of Corollary 3

The proof is derived from equation (24). When $\alpha \gg 1$, $q_{az} = \left(\frac{z - x_M}{x_H - z} \frac{\pi_2}{\pi_1} - 1 \right) (\delta + \gamma)$. As in Example 2, I assume that $q_{az} \in (0, q)$, which implies that $\frac{z - x_M}{x_H - z} \frac{\pi_2}{\pi_1} > 1$. Therefore, q_{az} is increasing in γ . When $\alpha \ll \delta$, $q_{az} = \left(\frac{z - x_M}{x_H - z} \frac{\pi_2}{\pi_1} - 1 \right) (\delta + \gamma) + \frac{z - x_M}{x_H - z} \frac{\pi_2}{\pi_1} q$. If $q_{az} \in (0, q)$, it must be that $\frac{z - x_M}{x_H - z} \frac{\pi_2}{\pi_1} < 1$. Therefore, q_{az} is decreasing in γ . Q.E.D.

A.10 Proof for Corollary 4

For δ_1, δ_2 such that $\delta_2 > \delta_1 > 0$, denote the trading frequency and cutoffs of δ_j by q_{axj} , $\bar{x}_j^a$ and $\underline{x}_j^a$. From equation (22) and (23), $\bar{x}_1^a = \bar{x}_2^a$, $\underline{x}_1^a < \underline{x}_2^a$. For all $x \in (\underline{x}_2^a, \bar{x}_2^a)$, $\frac{d}{dx}(q_{ax2} - q_{ax1}) = \frac{g(x)}{g(\bar{x}^a)} \left(\frac{x - x_M}{\bar{x}^a - x_M} \right)^2 (\delta_2 - \delta_1) > 0$. Since $q_{ax2} - q_{ax1} < 0$ at $\underline{x}_2^a$ and $q_{ax2} - q_{ax1} = 0$ at $x = \bar{x}_2^a$, $q_{ax2} < q_{ax1}$ for all $x \in (\underline{x}_2^a, \bar{x}_2^a)$. Q.E.D.

A.11 Proof for Corollary 5

From equation (24), $q_{az} = (\varphi - 1)(\delta + \gamma) + q \frac{\varphi}{1 - \frac{\varphi}{(\alpha + \delta)(\gamma + \delta)}}$. For q_{az} to be between 0 and q , $\varphi < 1$. Therefore, q_{az} is decreasing in δ . Q.E.D.

A.12 Proof for Corollary 6

The Corollary holds for Example 2, according to equation (24). For Example 1, consider ζ_1 and ζ_2 such that $\zeta_2 > \zeta_1 > 1$. Denote the trading frequency and cutoffs corresponding to ζ_j by q_{axj} , $\bar{x}_j^a$ and $\underline{x}_j^a$.

According to equation (22), $\bar{x}_2^a > \bar{x}_1^a$. According to equation (23), $\underline{x}_2^a > \underline{x}_1^a$ because

$$\underline{x}_2^a - x_M = \left(\frac{\delta}{\delta + q} \right)^{\frac{1}{2+\zeta_2}} (\bar{x}_2^a - x_M) > \left(\frac{\delta}{\delta + q} \right)^{\frac{1}{2+\zeta_1}} (\bar{x}_1^a - x_M) = \underline{x}_1^a - x_M.$$

For any $x \in [\underline{x}_2^a, \bar{x}_1^a]$, according to equation (21),

$$q_{ax2} = \left(\frac{x - x_M}{\bar{x}^a - x_M} \right)^{2+\zeta_2} (\delta + q) - \delta < \left(\frac{x - x_M}{\bar{x}^a - x_M} \right)^{2+\zeta_1} (\delta + q) - \delta = q_{ax1}.$$

Q.E.D.

A.13 Proof of Lemma 4

We focus without loss of generality on the ask contracts. By Lemma 3

$$F_a(R_{0x}) = q_{ax}/q, \quad f_a(R_{0x})R'_{0x} = q'_{ax}/q,$$

and

$$\begin{aligned}\int_{R_{0\kappa}}^{R_{0x}} R dF_a(R) &= p_{ax} q_{ax} / q, \\ q \int_{R_{0\kappa}}^{R_{0x}} R dF_a(R) &= (r + \delta + q_{ax} + \gamma) \left[P + \int_{\kappa}^x \frac{1 + \gamma e'_s}{r + \delta + q_{as} + \gamma} ds \right] - (x + \delta Ed + \gamma e_x), \\ q R_{0x} f_a(x) R'_{0x} &= q'_{ax} \left[P + \int_{\kappa}^x \frac{1 + \gamma e'_s}{r + \delta + q_{as} + \gamma} ds \right].\end{aligned}$$

Since $R'_{0x} f_0(R_{0x}) = q'_{ax} / q$, we have, $R_{0x} = P + \int_{\kappa}^x \frac{1 + \gamma e'_s}{r + \delta + q_{as} + \gamma} ds$. Since $d'_x = \frac{1 + \gamma \frac{1}{r + \alpha + \delta}}{r + \delta + q_{bx} + q_{ax} + \gamma \frac{r + \delta}{r + \alpha + \delta}}$ and $P = d_{\kappa}$, $P + \int_{\kappa}^x \frac{1 + \gamma e'_s}{r + \delta + q_{as} + \gamma} ds = d_{\kappa} + \int_{\kappa}^x d'_s ds = d_x$. Therefore, $R_{0x} = d_x = M_{1x} - M_{0x}$. Q.E.D.

A.14 Proof for Proposition 3

Under complete information, traders do not get any information rent,

$$\begin{aligned}(r + \delta + \gamma) d_x &= x + \delta Ed + \gamma e_x, \\ (r + \delta + \alpha) e_x &= x + \delta Ee + \alpha d_x, \\ Ed = Ee &= \frac{Ex}{r}, \\ P_0 = d_{\kappa} &= \frac{\kappa}{r + \delta} + \frac{\delta Ex}{r(r + \delta)}.\end{aligned}$$

Under asymmetric information, asset price also depends on the information rent extracted by traders,

$$\begin{aligned}P &= d_{\kappa} = \frac{1}{r + \gamma + \delta} (\kappa + \gamma e_{\kappa} + \delta Ed) \\ &= \frac{(r + \alpha)(r + \alpha + \delta + \gamma) \kappa + \gamma \delta Ex + [(r + \alpha)(r + \alpha + \delta) + \gamma \alpha] \delta Ed}{(r + \alpha)(r + \alpha + \gamma + \delta)(r + \delta)}.\end{aligned}\quad (38)$$

From equilibrium conditions, one can derive

$$Ed = \frac{Ex}{r} + \frac{(r + \alpha) \mathcal{I}}{r(r + \alpha + \gamma)}, \quad (39)$$

where

$$\begin{aligned}
\mathcal{I} &= - \int_{\kappa}^{x_H} \int_{\kappa}^x q_{as} d'_s ds dG(x) + \int_{x_L}^{\kappa} \int_x^{\kappa} q_{bs} d'_s ds dG(x) \\
&= - \int_{\kappa}^{x_H} \int_{\kappa}^x \frac{q_{as}}{r + \delta + \frac{r+\alpha+\delta}{r+\alpha+\delta+\gamma} q_{as}} ds dG(x) + \int_{x_L}^{\kappa} \int_x^{\kappa} \frac{q_{bs}}{r + \delta + \frac{r+\alpha+\delta}{r+\alpha+\delta+\gamma} q_{bs}} ds dG(x), \\
&= - \int_{\kappa}^{x_H} q_{ax} (d_x - p_{ax}) dG(x) + \int_{x_L}^{\kappa} q_{bx} (p_{bx} - d_x) dG(x).
\end{aligned}$$

Substituting equation (39) into equation (38) we have

$$P = \frac{\kappa}{r + \delta} + \frac{\delta Ex}{r(r + \delta)} + \frac{c\mathcal{I}}{r},$$

where $c = \frac{[(r+\alpha)(r+\alpha+\delta)+\gamma\alpha]\delta}{(r+\alpha+\gamma)(r+\alpha+\gamma+\delta)(r+\delta)}$. Q.E.D.

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